

Chat Bot for Telemetry Analysis on AROL Capping Machines

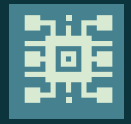
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The subject of our project



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AROL Capping Machines

Founded in 1978, **AROL** is a global provider of capping solutions.

AROL multi heads capping machines are capable of handling continuous, high-volume manufacturing, reaching outputs up to 100,000 bottles per hour. Providing model-specific ***metrics*** that are valuable for tracking performance, identifying anomalies, and facilitating predictive and preventive maintenance.

In this project, we concentrated on the ***36-head*** machine focusing on ***torque and closure diagnostics***.

Within the gathered ***data***, we observe the ***timestamp*** of the measurement and three metrics for each individual: ***Count***, ***AppTorque***, and ***Status***.



Multiheads Capping machine

L1. Ingestion and Preprocessing

Normalization Pipeline

1. Data **ingestion** - daily high volume
2. **Validate** schema (missing values)
3. Conversion into ***Parquet* format**
4. Recognition of significant **events**
5. Identification of **idle periods** (above 30 sec.)
6. **Join** data and **enrich** table (comment columns)
7. Handle **duplicates** (within segment)
8. **Export** data and report

timestamp	head_id	counter	torque_nm	status_code	status_label	is_successful	classification	capping_speed_pph
2026-02-07 06:38:30	H01	205088	2.014	0	Closure OK	True	successful	1525.4
2026-02-07 06:38:30	H02	198442	0.000	2	No Load	False	no_load	1521.2
2026-02-07 06:38:32	H29	210115	2.085	65	RotatingAtRaise	False	failed	1518.9
2026-02-07 06:38:33	H08	201304	2.012	0	Closure OK	True	successful	1528.1
2026-02-07 06:38:35	H24	209880	2.011	0	Closure OK	True	successful	1530.0

Enriched table

L2. Analytics functions

Metrics & Tools

1. **dataset_summary** - general overview (number of closures, number of heads, time range covered, and number of successful/failed/no-load cases, alert)
2. **success_rate_analysis** - success rate
3. **torque_statistics** - classic statistics on applied torque (mean, median, standard deviation, quartiles, outliers), consider only successful capping operations
4. **torque_trend_analysis** - trends in torque over time, per head (rising/falling/stable)
5. **anomaly_detection** - anomalous closures by pair (standard deviation, IQR, or a manual threshold) and time ranges with an anomalous failure rate.
6. **head_comparison** - one row per head (success rate, average pair value, variability, number of closings)

L2. Analytics functions

Metrics & Tools

7. **failure_analysis** - common error codes, frequent failures days, conjunct failures
8. **capping_speed_analysis** - consider only active hours
9. **idle_analysis** - usage rage, avg length, frequent inactive time range
10. **generate_kpi_dashboard** - overview dashboard
11. **list_events** - raw-line (filtered) list of individual closing events
12. **torque_outcome_comparison** - using Mann-Whitney U tests between successful closures and failures
13. **torque_success_correlation** - Pearson correlation between mean torque and success rate
14. **visualize** - renders chart as PNG image

L3. AI Agent

Interaction Pipeline

1. **Routing** - *router.route(question)*
   a. connect w/ LLM
   b. send system prompt and question
   c. convert answer in JSON (retry/enum normalization, if needed -> fallback)
   d. extract *tool_calls*

2. **Execution** - *executor.execute_all()*
   a. execute *tool_calls* and store results

3. **Composition** - *composer.compose()*
   a. combine together response as a text

4. **Reply** - *agent.query()*
   a. return *AgentResponse* obj

L4. Bot Interface

Alternative to CLI

Layer 3 offers CLI while layer 4 propose a **bot simulator**

Mode selection:

- *Guided mode* - through a menu accessing the 14 tools without use interaction with the LLM
- *Free mode* - direct chat with LLM, more similar to CLI

Graphs generation implemented, offering 6 models (torque_over_time, torque_histogram, success_rate_per_head, failures_over_time, production_over_time, utilization)

Execution Results Analysis

Ingestion of observed data:

- 4.6 GB in ~60 sec
- 89 days
- over 77 million rows of data
- over 55 million capping operations

AROL machine expectations:

- Throughput 50.000-72.000 pph

AROL AI Agents strengths:

- zero hallucination (hybrid architecture)
- reliability with unavailable connection
- double mode
- accurate over 35 queries

KPI	Value
Overall success rate (excl. no-load)	99.9965% (31,670,096 successful / 1,096 failed)
Mean torque (successful closures)	2.014 Nm
Torque stability (std of per-head means)	0.0010 Nm
Machine-wide throughput (1,212 hours with recorded closures)	26,610 pph
Per-head average speed	1,521.7 pph/head
Utilization rate (2137.0h observation window)	33.63%
Worst-performing head	H29 (99.99%)
Best-performing head	H24 (99.9995%)
Anomalies detected (z-score method)	307,821
Total idle time	1,418.3h

KPI dashboard

Demo



Thank you for the attention!

Do you have any questions?

